

Blueberry Supplementation Improves Memory in Older Adults

The prevalence of dementia is increasing with expansion of the older adult population. In the absence of effective therapy, preventive approaches are essential to address this public health problem. Blueberries contain polyphenolic compounds, most prominently anthocyanins, which have antioxidant and anti-inflammatory effects. In addition, anthocyanins have been associated with increased neuronal signaling in brain centers mediating memory function as well as improved glucose disposal, benefits that would be expected to mitigate neurodegeneration. We investigated the effects of daily consumption of wild blueberry juice in a sample of nine older adults with early memory changes. At 12 weeks, we observed improved paired associate learning ($p = 0.009$) and word list recall ($p = 0.04$). In addition, there were trends suggesting reduced depressive symptoms ($p = 0.08$) and lower glucose levels ($p = 0.10$). We also compared the memory performances of the blueberry subjects with a demographically-matched sample who consumed a berry placebo beverage in a companion trial of identical design and observed comparable results for paired associate learning. The findings of this preliminary study suggest that moderate-term blueberry supplementation can confer neurocognitive benefit and establish a basis for more comprehensive human trials to study preventive potential and neuronal mechanisms.